MARMARA UNIVERSITY

FACULTY OF ENGINEERING

PHYS 1104

PHYSICS LABORATORY II

Magnetic Field of Solenoids

Section: 12

Group: 5

Instructor: Magnetic Field of Solenoids Date: 02/05/2024 8:59 am

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1 Computer Engineering		
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2. Purpose

The aim of this experiment is to measure the magnetic flux density along the axis of a solenoid and experimentally determine the magnetic permeability of air. Utilizing Ampere's Law, measurements are conducted on solenoids with different numbers of turns and currents, comparing the results with theoretical values. This enables understanding the relationship between magnetic field and current, and the experimental determination of magnetic permeability in air.

3. Theory

The theoretical foundation of the experiment is based on one of the fundamental laws in electromagnetism, Ampere's Law. This law describes how the magnetic field inside a solenoid changes depending on the current passing through the solenoid and the geometric properties of the solenoid. According to Ampere's Law, the magnetic field intensity (**B**) inside a solenoid is related to the length of the solenoid (**L**), the current passing through it (**I**), and the number of turns in the solenoid (**N**), and can be calculated using the formula:

$B = (\mu_0 \cdot N \cdot I) / L$

Where:

- B represents the magnetic field intensity (in Tesla, T).
- μ_0 represents the magnetic permeability of free space (in T·m/A).
- N represents the number of turns in the solenoid.
- I represents the current passing through the solenoid (in Amperes, A).
- L represents the length of the solenoid (in meters, m).

The experiment aims to measure the magnetic field intensity using Ampere's Law and compare these measurements with theoretical values to experimentally determine the magnetic permeability of free space. By doing so, the goal is to better understand the relationship between magnetic field and current and evaluate the validity of experimental results by comparing them with theoretical predictions.

4. Experimental Setup

4.1 Equipment List

1. Circuit board x1
2. Digital multimeter x1
3. Cables
4. Gaussmeter with DC hall probe x1
5. G-clamp x1
6. Right angle clamp x1
7. Support rod x1

8. Ruler x1
9. 100 turns solenoids (L=0,043m)x1
10. 200 turns solenoids (L=0,090m)x1
11. 300 turns solenoids (L=0,143m)x1

4.2 Procedure

Part 1:

Preparation and Setup:

Gather the necessary equipment for the experiment and prepare a suitable area on the table to set up the apparatus.

Preparation of the Solenoid:

First, select the N1=100 turn solenoid and mount it onto the experimental setup.

Current Adjustment and Measurements:

Supply a constant current of 0.5 A through the solenoid using the circuit board.

Check and adjust the current using an ammeter connected in the circuit to ensure accuracy.

Position the Hall probe appropriately to measure the magnetic field intensity.

Magnetic Field Measurements:

Place the Hall probe at the entrance of the solenoid (z = 0) and start measuring the magnetic field intensity.

Move the Hall probe along the solenoid at specific intervals to measure the magnetic field intensity at different z values.

Record the measured values in a table and plot B values against z values.

Calculation of Magnetic Permeability:

Measure the magnetic field intensity at the center of the solenoid (z = L/2).

Calculate the experimental magnetic permeability (μ0) of the air using Ampere's Law.

Note the calculated value and perform a percentage error analysis by comparing it with the theoretical value.

Part 2:

Preparation and Setup:

Prepare the apparatus for using the 300-turn solenoid in the second part of the experiment.

Current and Magnetic Field Measurements:

Place the Hall probe at the midpoint of the solenoid.

Increase the current from 0 to 1.0 A in 10 steps and measure the magnetic field intensity at each step.

Record the measured values in a table and plot B values against current values.

Calculation of Magnetic Permeability:

Calculate the slope of the obtained graph.

Use the slope to calculate the experimental magnetic permeability (μ0).

Note the calculated value and perform a percentage error analysis by comparing it with the theoretical value.

Follow the steps carefully in both parts of the experiment, perform measurements accurately, and record the results.

5. Calculations and Results

5.1 Tables

Table 1: B values along z-axis of the solenoid

Z(cm)	**B(z) mT**		
	N_1=100	N_2=200	N_3=300
0	0.1	0.1	0.5
1.03	0.3	1.0	1.1
2.26	0.6	1.2	1.5
3.74	1.3	1.4	1.5
5.32	1.3	1.5	1.5
7.15	1.6	1.6	1.5
9.59	1.6	1.6	1.5
11.44	1.3	1.6	1.5
13.36	0.6	1.5	1.5
15.69	0.3	0.9	1.2

Table 2: Calculated value of μ_0

	N_1=100	N_2=200	N_3=300
H	13.5 mH	20.0 mH	30.0 mH
L	6.4 cm	9.0 cm	16.30 cm
B_0(mT)	1.6	1.6	1.5
μ_0	0.8×10^{-6}	0.9×10^{-6}	0.9×10^{-6}
%error			

Table 3: B values along current of the solenoid

I(A)	B (mT)
0	0
0.1	0.4
0.2	0.5
0.3	0.8
0.4	1.4
0.5	1.5
0.6	1.8
0.7	2.0
0.8	2.5
0.9	2.8
μ_0	1.0×10^{-6}
%error	10.5%

5.2 Calculations

The magnetic field inside a solenoid is given by:

$B = (\mu_0 \cdot N \cdot I) / L$

For each solenoid, we can calculate the magnetic permeability (μ_0) by rearranging:

$\mu_0 = (B \cdot L) / (N \cdot I)$

Using the measured magnetic field at the center of each solenoid, we can determine the experimental value of μ_0.

For the 100-turn solenoid:

- B = 1.6 mT = 1.6×10^{-3} T
- L = 6.4 cm = 0.064 m
- N = 100 turns

- I = 0.5 A
- $\mu_0 = (1.6 \times 10^{-3} \times 0.064) / (100 \times 0.5) \approx 0.8 \times 10^{-6}$ T·m/A

Similar calculations were performed for the 200-turn and 300-turn solenoids.

5.3 Graphs

[Note: Graphs would be inserted here showing B vs. z for different solenoids and B vs. I]

6. Discussion and Conclusion

The discrepancies between the experimental and theoretical results in the tables may stem from various factors. Measurement errors, material and structural properties, as well as limitations of the theoretical model, could contribute to the inconsistencies in the results.

From Table 2, we can see that the calculated values of μ_0 from all three solenoids are reasonably consistent, with values around 0.8-0.9 $\times 10^{-6}$ T·m/A. The theoretical value for the magnetic permeability of free space is $4\pi \times 10^{-7}$ T·m/A (approximately 1.26×10^{-6} T·m/A), which means our experimental values are somewhat lower than expected.

The third table shows a clear linear relationship between current and magnetic field as predicted by theory. The calculated μ_0 value from this part (1.0×10^{-6} T·m/A) is closer to the theoretical value with a 10.5% error.

Sources of error could include:

- Imprecise positioning of the Hall probe
- Variations in the solenoid winding density
- Measurement uncertainty in current and magnetic field readings
- Edge effects at the ends of the solenoid

Further investigation into these differences and conducting more precise measurements are warranted. Ultimately, more research is needed to better understand the role of magnetic fields in practical applications and to obtain more accurate results.

7. References

- https://canvas.instructure.com/files/249546919/download?download_frd=1
- https://canvas.instructure.com/files/245848696/download?download_frd=1

8. Appendix

Experiment Sheets

[Note: Original experiment sheets would be attached here]